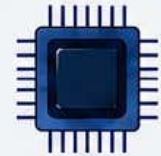
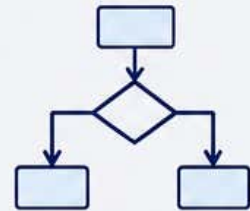




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Syllabus



Major Course (Elective-2 Laboratory): Laboratory on Machine learning

Level	Semester	Course Code	Course Name	Credits	Teaching Hours	Exam Duration	Max Marks
5.5	VI	112228	Laboratory on Machine learning	2	60	4Hrs	50

Course Objectives:	1. To familiarize students with Python libraries used for Machine Learning such as NumPy, Pandas, Matplotlib and Scikit-learn. 2. To develop practical skills for creating, loading, exploring and preprocessing datasets. 3. To enable students to apply data preprocessing techniques such as handling missing values, categorical encoding, feature scaling and train-test splitting. 4. To provide hands-on experience in implementing supervised and unsupervised Machine Learning algorithms. 5. To train students to evaluate Machine Learning models using standard performance measures.						
Course Outcomes:	On completion of the following syllabus the students will be able to - 1. Use Python libraries such as NumPy, Pandas, Matplotlib and Scikit-learn for Machine Learning practical. 2. Create, load, explore and preprocess datasets for Machine Learning applications. 3. Apply preprocessing techniques such as missing value handling, categorical data encoding, feature scaling and train-test splitting. 4. Implement supervised and unsupervised learning algorithms such as Linear Regression, Logistic Regression, KNN, Decision Tree, K-Means and PCA. 5. Evaluate Machine Learning models using confusion matrix, accuracy, precision and recall.						
Contents					Workload Allotted	Weightage of Marks Allotted	Incorporation of Pedagogies
	List of Practical: 1. Study and use NumPy, Pandas, Matplotlib and Scikit-learn for Machine Learning. 2. Create and load a dataset using Pandas. 3. Perform data preprocessing by handling missing values. 4. Apply categorical data encoding and feature scaling. 5. Split a dataset into training and testing data. 6. Implement Simple Linear Regression. 7. Implement Logistic Regression for classification. 8. Implement K-Nearest Neighbours classification. 9. Implement Decision Tree classification.						Use any pedagogical technique suitable for the topic

	10. Implement K-Means clustering and plot clusters. 11. Apply PCA for dimensionality reduction. 12. Evaluate a classification model using confusion matrix, accuracy, precision and recall.			
References:	Web-link to Equivalent MOOC on SWAYAM if relevant: https://nptel.ac.in/courses/106106179 https://nptel.ac.in/courses/106105152 https://onlinecourses.nptel.ac.in/e-learning/preview/noc20_cs44 https://nptel.ac.in/courses/106106179			

Assessment Rubric: External Practical Course

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI PRACTICAL EXAMINATION B.Sc. III, SEMESTER – VI (NEP)				
Practical	Course Code: 112228	Laboratory on Machine learning	Max Marks: 25	Time: 4 Hrs.
Sr. No.	Exercise			Marks
1.	Single Program Writing, Algorithm, Flowchart, Execution, with Sample inputs and Outputs, and Printout			10
2.	Submitted Record book (Preferably handwritten)			05
3.	External Viva Voce			10

Assessment Rubric: Internal Practical Course

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI PRACTICAL EXAMINATION B.Sc. III, SEMESTER – VI (NEP)			
Practical	Course Code: 112228	Laboratory on Machine learning	Max Marks: 25
Sr. No.	Assessment Criteria		Marks
1.	Attendance/Regularity in practical's / Students' overall performance		05
2.	Record book Preparation (Preferably handwritten)		10
3.	Internal Viva Voce		10